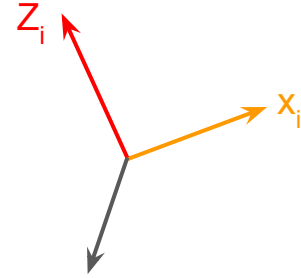
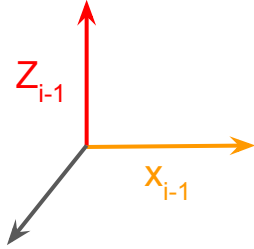


Robotics Kinematics

BRAC University

How to Represent the Transform between 2 frames



Here, the relative relationship is found by the four DH parameters

How to Represent the Transform between 2 frames

We need a representation that allows:

- chaining transformations
- applying transformations to points
- efficient computation

Homogeneous Matrix

Homogeneous transformation allows us to represent **rotation and translation** in a single matrix, enabling simple and **efficient chaining** of robot motion.

DH parameters describe **geometry**. Homogeneous matrices make it **computable**.

DH Transformation Matrix for a single joint

- Therefore, the transformation is in the reverse order:

$${}^{i-1}T = T_z(\theta_i)T_z(d_i)T_x(a_i)T_x(\alpha_i)$$

- Thus, the DH transformation matrix:

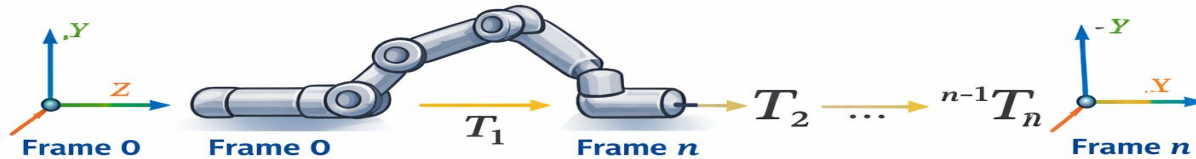
$$\begin{bmatrix} \cos \theta_i & -\sin \theta_i \cos \alpha_i & \sin \theta_i \sin \alpha_i & a_i \cos \theta_i \\ \sin \theta_i & \cos \theta_i \cos \alpha_i & -\cos \theta_i \sin \alpha_i & a_i \sin \theta_i \\ 0 & \sin \alpha_i & \cos \alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

[Visualization](#) Link

Manipulator Transformation Matrix

The position and orientation of end effector frame relative to the based frame can be found by consisting n consecutive link transformation matrices relating frames fixed to adjacent links.

$$T = {}^0T_n = {}^0T_1 {}^1T_2 \dots {}^{n-1}T_n$$

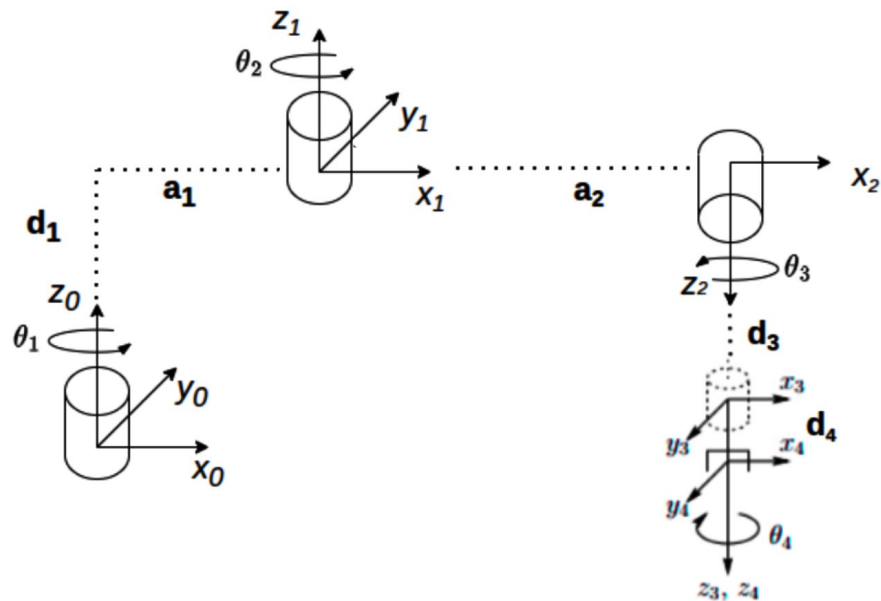


Recall the Definition of the Four Parameters

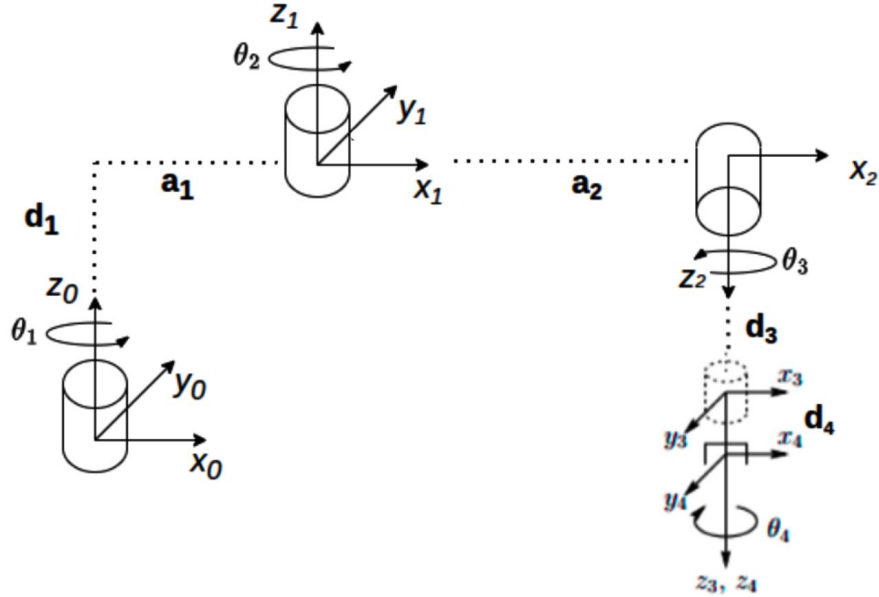
1. Link Length (a_i): Distance between z_{i-1} and z_i along x_i
2. Link Twist (α_i): Angle between z_{i-1} and z_i about x_i
3. Joint Offset (d_i): Distance between x_{i-1} and x_i along z_{i-1}
4. Joint Angle (θ_i): Angle between x_{i-1} and x_i about z_{i-1}

Note: Don't try to imagine the Y-axis

Exercise 1



Exercise 1



i	Θ_i	a_i	d_i	α_i
1	$+\Theta_1$	a_1	d_1	0
2	$+\Theta_2$	a_2	0	180°
3	$+\Theta_3$	0	d_3	0
4	$+\Theta_4$	0	d_4	0

Exercise 1 (Cont.)

$$T_i = \begin{bmatrix} \cos\theta_i & -\cos\alpha_i\sin\theta_i & \sin\alpha_i\sin\theta_i & a_i\cos\theta_i \\ \sin\theta_i & \cos\alpha_i\cos\theta_i & -\sin\alpha_i\cos\theta_i & a_i\sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

i	Θ_i	a_i	d_i	α_i
1	$+\Theta_1$	a_1	d_1	0
2	$+\Theta_2$	a_2	0	180°
3	$+\Theta_3$	0	d_3	0
4	$+\Theta_4$	0	d_4	0

Exercise 1 (Cont.)

T_{01}

$$T_{01} = \begin{bmatrix} c_1 & -s_1 & 0 & a_1 c_1 \\ s_1 & c_1 & 0 & a_1 s_1 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

T_{12} (note: $\alpha_2 = \pi \Rightarrow \cos \alpha_2 = -1, \sin \alpha_2 = 0$)

$$T_{12} = \begin{bmatrix} c_2 & s_2 & 0 & a_2 c_2 \\ s_2 & -c_2 & 0 & a_2 s_2 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

i	Θ_i	a_i	d_i	a_i
1	$+\Theta_1$	a_1	d_1	0
2	$+\Theta_2$	a_2	0	180°
3	$+\Theta_3$	0	d_3	0
4	$+\Theta_4$	0	d_4	0

Exercise 1 (Cont.)

T_{23}

$$T_{23} = \begin{bmatrix} c_3 & -s_3 & 0 & 0 \\ s_3 & c_3 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

T_{34}

$$T_{34} = \begin{bmatrix} c_4 & -s_4 & 0 & 0 \\ s_4 & c_4 & 0 & 0 \\ 0 & 0 & 1 & d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

i	Θ_i	a_i	d_i	a_i
1	$+\Theta_1$	a_1	d_1	0
2	$+\Theta_2$	a_2	0	180°
3	$+\Theta_3$	0	d_3	0
4	$+\Theta_4$	0	d_4	0

Exercise 1 (Cont.)

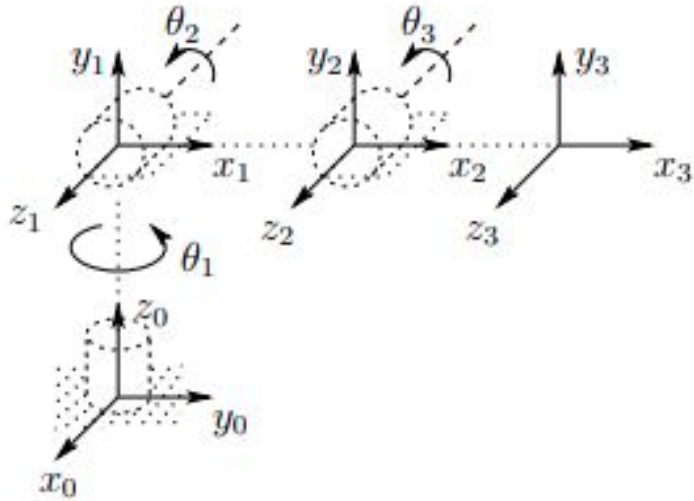
$$T_{04} = T_{01}T_{12}T_{23}T_{34}$$

$$T_{04} = \begin{bmatrix} \cos \psi & \sin \psi & 0 & x \\ \sin \psi & -\cos \psi & 0 & y \\ 0 & 0 & -1 & z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Where:

- $\psi = \theta_1 + \theta_2 - \theta_3 - \theta_4$
- $x = a_1 \cos \theta_1 + a_2 \cos(\theta_1 + \theta_2)$
- $y = a_1 \sin \theta_1 + a_2 \sin(\theta_1 + \theta_2)$
- $z = d_1 - d_3 - d_4$

Exercise 2



i	Θ_i	a_i	d_i	α_i
1	$+\Theta_1$	0	h	90°
2	$+\Theta_2$	L_2	0	0
3	$+\Theta_3$	L_3	0	0

Exercise 2 (Cont.)

$$T_i = \begin{bmatrix} \cos\theta_i & -\cos\alpha_i \sin\theta_i & \sin\alpha_i \sin\theta_i & a_i \cos\theta_i \\ \sin\theta_i & \cos\alpha_i \cos\theta_i & -\sin\alpha_i \cos\theta_i & a_i \sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

i	Θ_i	a_i	d_i	α_i
1	$+\Theta_1$	0	h	90°
2	$+\Theta_2$	L_2	0	0
3	$+\Theta_3$	L_3	0	0

Calculate 0T_1 , 1T_2 , 2T_3

Exercise 2 (Cont.)

$${}^0T_1 =$$

$${}^1T_2 =$$

Calculate ${}^0T_3 = {}^0T_1 \times {}^1T_2 \times {}^2T_3$

$${}^2T_3 =$$

Exercise 2 (Cont.)

$0T_1$

For $\alpha_1 = 90^\circ \Rightarrow \cos \alpha_1 = 0, \sin \alpha_1 = 1, a_1 = 0, d_1 = h$:

$${}^0T_1 = \begin{bmatrix} c_1 & 0 & s_1 & 0 \\ s_1 & 0 & -c_1 & 0 \\ 0 & 1 & 0 & h \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$2T_3$

($\alpha_3 = 0, d_3 = 0, a_3 = L_3$):

$1T_2$

($\alpha_2 = 0, d_2 = 0, a_2 = L_2$):

$${}^1T_2 = \begin{bmatrix} c_2 & -s_2 & 0 & L_2c_2 \\ s_2 & c_2 & 0 & L_2s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2T_3 = \begin{bmatrix} c_3 & -s_3 & 0 & L_3c_3 \\ s_3 & c_3 & 0 & L_3s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

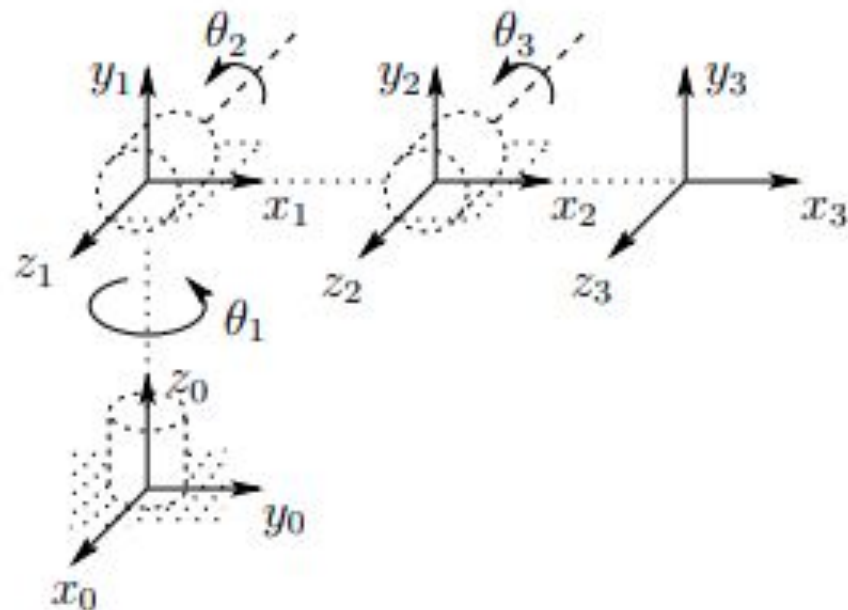
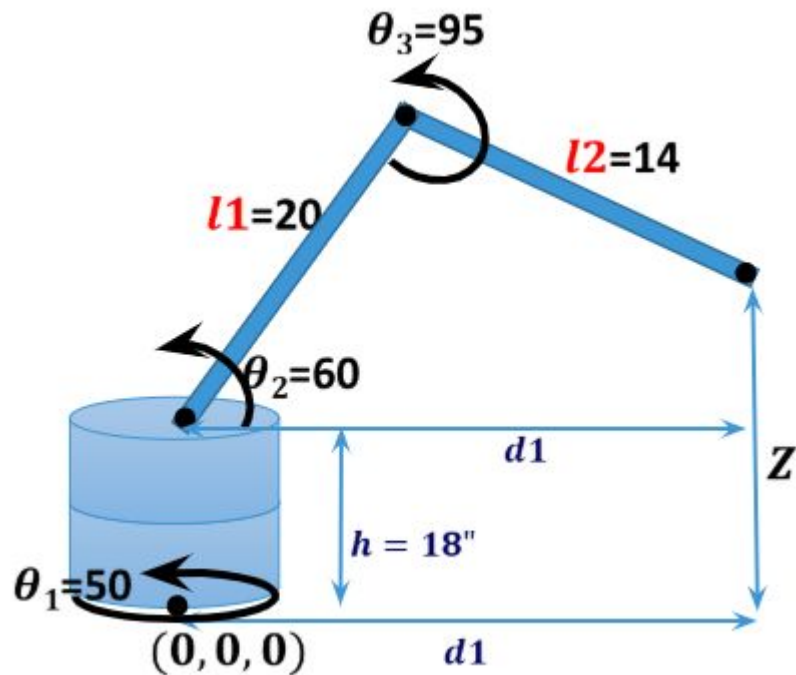
Exercise 2 (Cont.)

Final transform ${}^0T_3 = {}^0T_1 {}^1T_2 {}^2T_3$

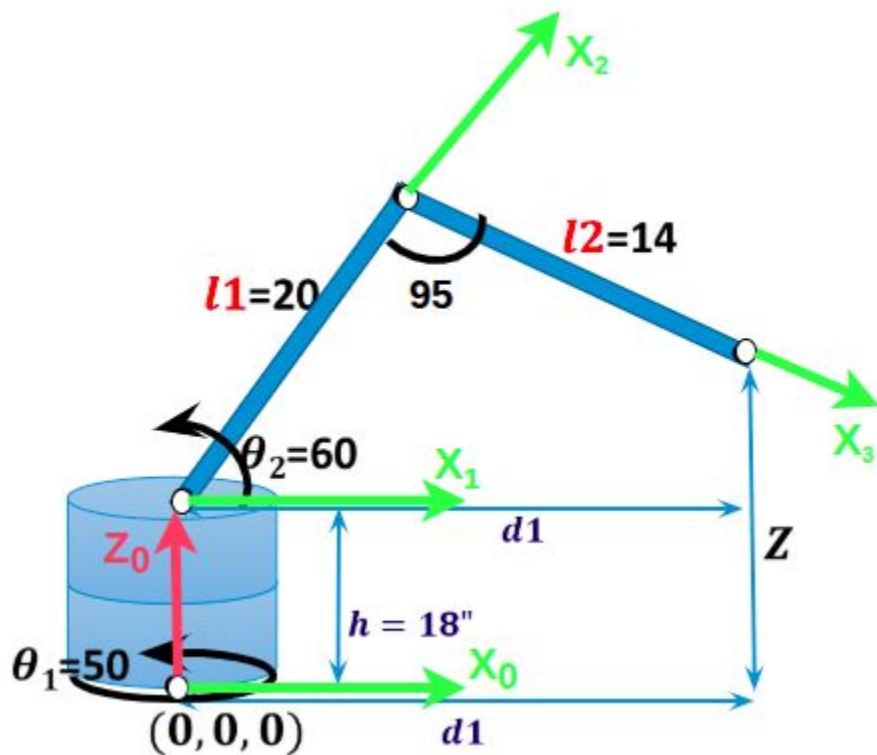
$${}^0T_3 = \begin{bmatrix} c_1 c_{23} & -c_1 s_{23} & s_1 & c_1 (L_2 c_2 + L_3 c_{23}) \\ s_1 c_{23} & -s_1 s_{23} & -c_1 & s_1 (L_2 c_2 + L_3 c_{23}) \\ s_{23} & c_{23} & 0 & h + L_2 s_2 + L_3 s_{23} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Did you familiar with the x, y and z value from the trigonometry method

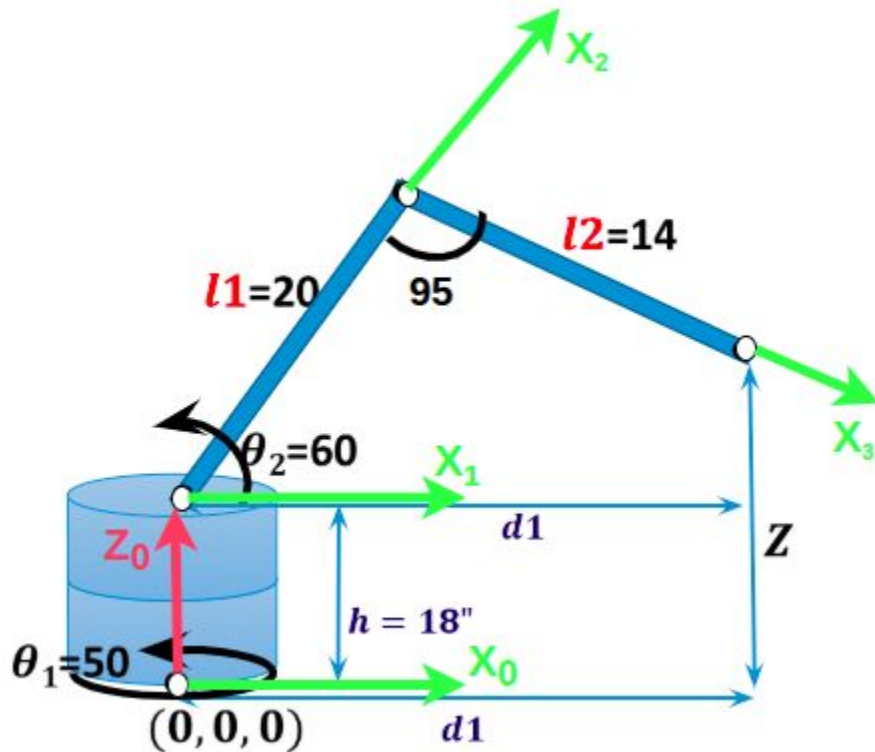
Home Work



Home Work (Cont.)



Home Work (Cont.)



Joint				
1	0+50	90	0	18
2	0+60	0	20	0
3	0-85	0	14	0

Home Work (Cont.)

Joint				
1	0+50	90	0	18
2	0+60	0	20	0
3	0-85	0	14	0

$$(X, Y, Z) = (14.584, 17.38, 29.404)$$

$$\begin{bmatrix} 0.583 & 0.272 & 0.766 & 14.584 \\ 0.694 & 0.324 & -0.643 & 17.38 \\ -0.423 & 0.906 & 0 & 29.404 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$